Sahaj Bhadja

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Summary

MSCS candidate in Los Angeles with internship experience writing production **C++** and **Python** code, building simulation and test infrastructure, **CI/CD** pipelines, and fault-isolation patterns with strong documentation and cross-functional collaboration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, OOP, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built production **Python** services on **FastAPI** and **PostgreSQL** with **CI/CD**, automated validation, and structured logging for 500+ users while improving latency **1.2s to 350ms** and maintaining 99%+ reliability.
- Implemented fault-tolerant async pipelines with job-state transitions, failure isolation, and reconciliation checks processing 5K+ monthly jobs with documented requirements and test-backed release workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed production **Python** services on **AWS** with **GitHub Actions CI/CD** and unit validation for 10K+ daily requests, owning documentation and stable performance under peak load with cross-functional partners.
- Contributed integration and batch testing for concurrent workflows, reducing feedback turnaround **60%** through code reviews while independently debugging production failures and improving maintainability.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered cross-platform software with **Jenkins CI/CD**, nightly integration testing, and 20+ **PyTest**/Mockito unit suites across **Linux**, Windows, and macOS, boosting pipeline efficiency **68%** in Agile sprints.
- Documented deployment workflows and optimized API integrations, improving response times **30%** while collaborating with Microsoft/GitHub stakeholders on verification and validation activities.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Automated **Docker** and **Jenkins CI/CD** pipelines with integration testing for eight microservices, cutting deployment time **50%** while improving observability through structured logging and error tracking.
- Built data sync for 2M+ records on **Azure SQL** with schema design and query tuning, strengthening reliability for high-volume processing with documented system behavior and test validation.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **Flask API** systems for 50K+ records with sub-200ms responses across 12+ endpoints, documenting designs and collaborating on integration requirements for cross-platform delivery workflows.
- Diagnosed production defects via logging and monitoring, reducing manual handling **70%** while supporting automated validation and webhook-driven processing across distributed system components.

Skills

Languages: C++, Python, Java, Bash, SQL, JavaScript
Systems: operating systems, algorithms, OOP, Linux, performance optimization
Testing: PyTest, unit testing, integration testing, validation, automated testing, debugging
Reliability: fault isolation, logging, monitoring, documentation, CI/CD, Git
Practices: Jenkins, GitHub Actions, Docker, AWS, Azure, Agile, cross-functional collaboration

Projects

CPU-scheduling Simulator Live | *C++, Simulation, Operating Systems*

- Built system simulation tooling implementing FCFS, SJF, SRT, RR, priority, and MLQ scheduling in **C++** with edge-case validation across 1K+ process combinations and dynamic metric reporting.

Linguistix Code | *Python, AWS ECS, Docker, CI/CD*

- Developed containerized processing tooling on **AWS ECS/Fargate** with **Docker** and CodePipeline **CI/CD**, automated validation gates, and production test infrastructure across deployment pipelines.

Chat2SQL Code | *Java, TypeScript, Docker, Validation*

- Developed schema-aware SQL transformation tooling with validation pipelines across 10+ database systems on **Java** and **TypeScript**, supporting cross-platform **Docker** deployment on Windows, macOS, and **Linux**.